# How Do LLM Respondents Use Demographics? Prompting, Fine-Tuning and Held-Out-Item Transfer for Survey Simulation in India

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***Abstract*: Silicon sampling asks a language model to answer a survey as if it were a respondent with a given demographic profile, which would be far cheaper than fielding a questionnaire. We ask what the profile actually contributes and what fine-tuning changes, using Pew Research Center's Religion in India survey (N = 29,999) and the World Values Survey wave 7 (WVS-7) for India (N = 1,692), with every method scored on the same respondent-item rows as matched classifiers. On a 200-respondent Pew subset, zero-shot Qwen2.5-7B-Instruct gets 17.5% right, against 30.2% for always giving the most common answer. It reacts strongly to the profile (the profile explains 29% of its prediction variance, against 4% for real answers), but its group differences correlate only r = 0.22 with the real ones, and removing the profile raises accuracy by 3.6 points. After fine-tuning with QLoRA (quantised low-rank adaptation) on 184,853 respondent-level examples, the model reaches 38.0% on all 6,000 held-out respondents (majority 30.4%, logistic regression 38.5%, gradient boosting 40.0%) and is well calibrated (expected calibration error 0.009). The gain did not carry over to new questions in our test: an approximate port of the adapter scores 28.9% on eight unseen items against 28.5% for its base model. The in-domain gain is 13.7 points larger over respondents, but with eight items per side that difference cannot be told apart from zero. Fine-tuning took about 6.6 GPU-hours, scoring costs about 0.2 s per answer on a T4 GPU, and a classifier is orders of magnitude cheaper. A 120B zero-shot model scores 27.6% on 100 respondents, below the 30.0% majority class (not significant). Logistic regression needs about 400 labelled respondents, and gradient boosting about 1,600, to clear the majority class. On WVS-7 India, three zero-shot large language models (LLMs) score 22 to 26% on 15 items where the majority class scores 21 to 25%, and their group differences carry no directional signal.***

***Index Terms*: silicon sampling, survey simulation, large language models, parameter-efficient fine-tuning, QLoRA, algorithmic fidelity, model calibration, demographic conditioning, computational social science, Indian public opinion, Pew Research Center, World Values Survey**

## I. Introduction

Silicon sampling asks a language model to answer a survey question as a respondent with a given demographic profile would. If it worked, it would be a cheap complement to polling [1]. So far the evidence is mixed. Models reproduce demographic-attitude correlations in US data [1] but misrepresent many groups [2]. At the individual level they match or trail statistical baselines trained on the same demographics [3], [4], over-attribute attitudes to demographics and inflate gaps between groups [3], and persona variables explain little of the real variance [5].

What these benchmarks leave open is what the profile itself contributes. Does a prompted model use the demographic information in the right direction? What does fine-tuning on respondent-level data change? And does that carry over to questions the model never saw, the one case where a classifier fitted per item has no labels to learn from?

We study these questions in India, using Pew's Religion in India survey and wave 7 of the World Values Survey. Four controls separate the profile's effect from the model's prior: a prompt with no profile, a label-free prior correction, a fine-tuned copy of the same model, and a test on held-out items. The questions are: how a prompted LLM uses the profile (RQ1); what fine-tuning changes relative to the zero-shot model and to matched classifiers (RQ2); whether the adaptation transfers to unseen questions (RQ3); and whether a larger zero-shot model or more labelled respondents change the picture (RQ4).

We contribute (i) profile-use diagnostics that separate accuracy from mechanism: a profile ablation, the demographic variance share, and the directional validity of group differences; (ii) evidence from India that a zero-shot 7B model uses the profile heavily but with weak direction, while fine-tuning on respondent-level data recovers most of the gap to matched classifiers; (iii) a held-out-item test, reported with its limits, in which the adapter shows no detectable gain on

unseen questions; and (iv) checks on model scale, on how many labelled respondents classical models need, and a WVS-7 replication.
Section II reviews related work, Section III describes the data, Section IV presents the evaluation framework, Section V reports computational requirements and scalability, Section VI gives the results, and Sections VII to IX discuss the findings, limitations and future work.

## II. Related Work

**Simulating respondents.** Conditioning a model on a sociodemographic backstory can reproduce demographic-attitude correlations in US data [1], but misalignment with many groups persists after steering [2], opinions resemble some countries more than others [6], and variance shrinks [7]. Persona variables explain under 10% of annotation variance [5], and minority religious viewpoints in Asia are poorly represented [8].
**Baselines and adaptation.** Random forests perform comparably to LLMs at the individual level on the American National Election Studies, with LLMs matching aggregates better [4]. Across the General Social Survey and the 63-country WVS, no model beats the strongest baseline, models over-determine attitudes by demographics, and segment gaps are inflated two- to fourfold [3]. Recent reports find negligible individual-level recovery on the European Social Survey [9] and in Singapore [10], and demographic information that is readable inside a 7B model but barely used [11]. Fine-tuning on survey data generalises to unseen subpopulations and surveys [12], partly to unseen questions and countries [13], and helps with small pilot samples [14]; rectification can be worth more than fine-tuning when human data are scarce [15], and respondent embeddings can be trained over a frozen model [16]. We take the baseline and over-determination findings as established. What we add is an India study with respondent-level fine-tuning and a same-model zero-shot control, diagnostics that separate how much the profile is used from whether it is used in the right direction, and a held-out-item test.

## III. Datasets

### A. Pew Religion in India

The Pew Research Center survey Religion in India [17] is a face-to-face survey of 29,999 adults fielded between November 2019 and March 2020. We parsed question wording and labels from the survey's DDI metadata; 73 items passed screening (binary items, demographic columns, items with more than 10% missing answers and near-unanimous items were excluded). The first eight items of the screened list (religious practice and identity, norms on politics and abortion, sex-selective practice, number of children; 4 to 8 answer options) are used for fine-tuning. Items 9 to 16 (meditation, self-rated health, trust in health advice, what it takes to be “truly Indian”, television serials, praying) are never used for training and form the transfer set. Respondents are split into five folds stratified by urban/rural and income band, and this paper uses fold 0: 23,999 training and 6,000 test respondents, giving 46,209 test answers on the trained items. The prompt carries nine attributes: sex, age band, marital status, education, religion, caste, urban/rural, region and income band.

### B. WVS-7 India

The World Values Survey wave 7 for India (version 6.0 [18]; N = 1,692) was screened against its codebook from 373 candidate items down to 144. The zero-shot runs use 15 of them and a 14-attribute prompt.

## IV. Proposed Evaluation Framework

Fig. 1 shows the framework. Every predictor receives the same prompt and is scored on the same respondent-item rows, so differences between methods cannot come from different data.

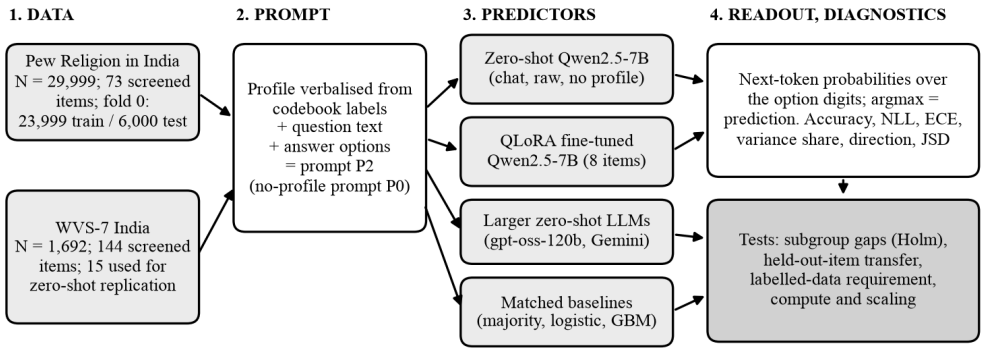



Fig. 1. Evaluation framework. Both surveys are verbalised into the same prompt, passed to zero-shot, fine-tuned, larger and classical predictors, and scored on identical rows with the same diagnostics and tests.

### A. Prompt and Readout

Each respondent-item pair becomes one prompt (P2): a sentence listing the respondent's demographic profile, the question text, the numbered answer options and an instruction to reply with a single digit (Table I). The profile is verbalised from codebook labels (nine attributes for Pew, 14 for WVS-7). The model's next-token distribution over the option digits gives the probabilities, and its argmax gives the hard prediction. The same prompt without the profile (P0) isolates the profile's contribution.

TABLE I. EXAMPLE PROMPT (P2) FOR ONE RESPONDENT AND ONE PEW ITEM.

| |
|---|
| You are a survey respondent with the following demographic profile: Sex: Female. Age: 18-25. Marital status: Married or living with a partner. Religion: Muslim. Urban/Rural: Rural. Region: East. Caste: General Category.<br>As this person, please answer the following survey question. Your answer should reflect the values and perspectives typical of someone with your background:<br>Question: How much influence should politicians have in religious matters?<br>Answer options: 1. A large influence 2. Some influence 3. Not too much influence 4. No influence at all<br>Your answer:<br>This is anonymized survey-simulation research. Respond with exactly one character: one of 1/2/3/4. No words, no punctuation, no explanation, just that single digit. |

### B. Baselines and Controls

**Matched baselines.** For each item we fit, on the training respondents and with exactly the prompt's attributes, a majority class, a logistic regression and a histogram gradient-boosting model (GBM) [19], plus a six-attribute cell mode for WVS. All are scored on the same rows as the LLMs.

**Zero-shot model.** Qwen2.5-7B-Instruct [20] was run locally for the 200-respondent subset (4-bit MLX [21]) and, for the full fold, on a Kaggle T4 GPU (NF4, fp16). The next-token distribution over the option digits gives both a hard prediction and probabilities. We use four conditions: *chat* (chat template), *raw* (bare prompt, the fine-tuning format), *no profile* (the same prompt for every respondent) and *prior-corrected* (probabilities rescaled by the training marginal over the model's own mean prediction, which needs no test labels).

**Larger zero-shot model.** gpt-oss-120b [22] (free tier, low reasoning effort) was run on a subset of respondents with the same prompts.

### C. Fine-Tuned Model

We used QLoRA [23], [24]: 4-bit NF4 weights, rank 16, alpha 32, dropout 0.05 on the q/k/v/o projections (10.1M trainable parameters). Training ran for one epoch over 184,853 examples from 23,999 respondents, with effective batch 32, peak learning rate 2e-4, seed 42 and loss over the whole sequence, and took 6.6 hours on one GPU. Predictions are the argmax of the next-token distribution over the option digits after the training-format prompt, scored on a Kaggle T4 in fp16. We use the same readout for every zero-shot condition. The original run decoded greedily and ended the prompt without the separator; that readout scores 36.1% and agrees with ours on 71.7% of rows. We report both and treat the training-format readout as primary. The transfer test in Section VI-C used an MLX re-implementation of the adapter on a 4-bit MLX base, which is only approximate.

### D. Diagnostics

All use answer indices normalised to [0, 1] on ordinal items. The *demographic variance share* is the increase in R-squared when profile dummies are added to item fixed effects, computed for predictions and for true answers; it parallels the variance-explained index in [3]. *Directional validity* works on (item, axis, category) cells with enough respondents (15 on the subset, 100 on the full Pew fold, 8 on WVS-7). For each cell we take the category mean minus the item mean, for predictions and for the truth, and report the Pearson correlation r between the two and the slope of predicted on true (below 1 is attenuation, above 1 inflation). Intervals resample respondents. Resampling attenuates cell means, so the intervals sit below the point estimates and show uncertainty, not exact coverage. The variance share uses probability-weighted predictions for the zero-shot model and hard predictions for the fine-tuned model and GBM. *Distributional fidelity* is the Jensen-Shannon divergence (base 2) between the mean predicted probability vector and the empirical answer distribution in each (item, subgroup) cell with at least 40 respondents, averaged over cells. We also report exact-match accuracy, negative log-likelihood (NLL), expected calibration error (ECE), paired respondent-bootstrap differences, and Holm-corrected permutation tests for subgroup accuracy gaps (best minus worst category).

### E. Transfer and Data-Efficiency Protocols

**Transfer.** The adapter and its base are applied to the eight unseen items for 120 respondents. Labels are used only for scoring. Classifiers trained with labels on those items give a ceiling that is not available for a genuinely new question. The in-domain gain is computed on the same respondents, and the difference of the two gains is bootstrapped over respondents. **Labelled-data requirement.** Classical models are trained on nested respondent subsets (25 to 23,999 respondents, five random subsets each) and scored on the same 200 test respondents, which gives the number of labelled respondents needed to clear the majority class.

## V. Computational Requirements and Scalability

### A. Measured Cost

Table II lists the hardware and measured cost of each part of the study. All figures come from our own runs. Scoring one condition on the full Pew fold (46,209 rows) took about 75 minutes on two Kaggle T4 GPUs working in parallel, which is about 0.19 to 0.21 s per row per GPU at batch size 16 and a mean prompt length of 231 tokens (maximum 288). The fine-tuning run took 5,777 optimizer steps and 396.7 minutes on a single lab GPU. In a separate reduced run on a T4, an optimizer step over 16 examples took 9 to 10 s, which is about 0.6 s per example. The 4-bit base model fits in about 4 to 5 GB of weights, so the 7B model runs on a 16 GB laptop or a 15 GB T4.

Table II. Hardware and measured cost.

| Task | Hardware | Measured cost |
|---|---|---|
| QLoRA fine-tuning, 184,853 examples | One lab GPU | 5,777 steps, 6.6 h (396.7 min) |
| Scoring one condition, 46,209 rows | Two Kaggle T4, fp16 | About 75 min; 0.19-0.21 s per row per GPU |
| Local zero-shot scoring | Apple M1, 16 GB, 4-bit MLX | 2.3 s per prompt with prefix caching; about 5 h per 1,000 respondents |
| Matched classical baselines, 8 items | Laptop CPU | Under 1 min for the whole fold |
| gpt-oss-120b, 774 rows | Free API tier | Limited by rate limits; several hours |

### B. Complexity and Scaling

Cost grows linearly with the number of respondent-item rows, because every row needs one forward pass over a prompt of about 230 tokens, which is roughly 2 x 7.6 billion x 230, or 3.5 TFLOP, for a 7B model. Training cost grows linearly with the number of examples. It does not depend on the number of subgroups. Scoring all 29,999 Pew respondents on all 73 screened items (about 2.2 million rows) would take roughly 116 GPU-hours on a T4 at the measured rate, or about 58 hours on two. Fine-tuning on all 73 items (about 1.75 million examples) would take roughly 63 hours at the lab GPU's rate and about 290 hours at the T4 rate. These are linear extrapolations, not measurements. A classifier, by contrast, trains in seconds per item and predicts a row in microseconds, several orders of magnitude cheaper.

### C. Practical Implications

Because the fine-tuned model is no more accurate than matched classifiers on the trained items (Section VI-B), its cost has to be justified by something else, such as transfer to new questions or use as a generator of full response profiles. Our transfer test did not show such a benefit. Cost can be cut by caching the profile prefix across a respondent's items, as our local runs did, by batching, and by using smaller or distilled models, but we did not test the last of these.

## VI. Results and Analysis

### A. RQ1: The Profile Is Used Heavily, Not Usefully

Table III summarises the zero-shot behaviour. With the chat template the model scores 17.5% (95% CI 15.7 to 19.4), 12.7 points below the majority class and at chance (19.4%). The raw prompt does the same (18.8%, 120 respondents). Removing the profile raises accuracy to 21.1%, so the profile changes accuracy by -3.6 points (-5.9 to -1.2). The model still reacts strongly to it: its prediction changes on 50% of rows relative to no profile, and the profile explains 29% of its prediction variance, against 4% for real answers and 6% for GBM. Its group differences are weakly aligned with the truth ($r = 0.22$, 95% CI 0.02 to 0.35) and strongly attenuated (slope 0.18), as Fig. 2 shows. On the full fold (6,000 respondents, 46,209 rows) the same model scores 16.1% with the chat template (95% CI 15.7 to 16.4) and 18.0% with the raw prompt, against 19.4% for chance and 30.4% for the majority class, so the subset result is not a sampling artefact. A prior correction recovers about two thirds of the distance to the majority class (26.3%) and no more. The model's NLL (3.88) is worse than guessing uniformly, with probability concentrated on one option for most respondents (mean top-option share 0.71).

Table III. Pew, 200 held-out respondents (1,548 rows). Direction and variance columns exclude one non-ordinal item (1,351 rows, 182 cells). Dir. r: correlation between predicted and true group differences. Profile var.: variance explained by the profile beyond item effects.

| Predictor (200 respondents) | Acc. (%) [95% CI] | NLL | Dir. r | Profile var. |
|---|---|---|---|---|
| True answers | - | - | - | 4.0% |
| Majority class (uses labels) | 30.2 [28.1, 32.4] | 1.59 | - | - |
| Zero-shot, no profile | 21.1 [19.1, 23.1] | 3.92 | - | 0.0% |
| Zero-shot, profile (chat) | 17.5 [15.7, 19.4] | 3.88 | 0.22 | 29.4% |
| Zero-shot, prior-corrected | 26.3 [24.0, 28.6] | - | - | - |
| Fine-tuned (training-format readout) | 37.7 [35.1, 40.3] | - | 0.58 | 7.1% |
| Gradient boosting (uses labels) | 40.4 [37.3, 43.4] | 1.43 | 0.64 | 6.1% |

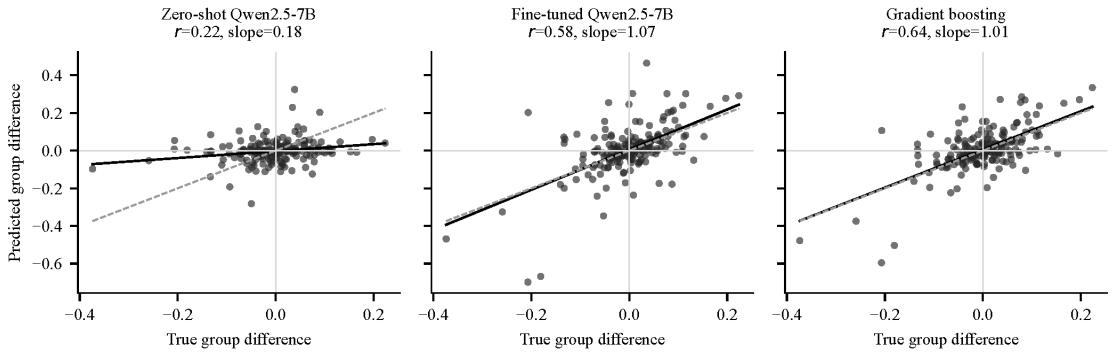



Fig. 2. Predicted versus true group differences for 182 (item, axis, category) cells (200 respondents; item-centred, normalised scale). The dashed line is the identity. Zero-shot differences are nearly flat and weakly correlated; the fine-tuned and GBM predictions follow the diagonal.

## *B. RQ2: Fine-Tuning Recovers Most of the Gap, Short of GBM*

On the full fold (Table IV) the fine-tuned model reaches 38.0% (95% CI 37.5 to 38.4). That is 7.6 points above the majority class, 0.5 below logistic regression (-0.8 to -0.2) and 2.1 below GBM, so it closes 79% (75 to 83) of the gap between the majority class and GBM. Its NLL of 1.48 sits between logistic regression (1.47) and the majority class (1.59) and above GBM (1.44), and it is well calibrated (ECE 0.009). It trails GBM on every item, by 1.2 to 2.7 points. Averaging its probabilities with GBM's gives no accuracy gain over GBM (-0.2 points, 95% CI -0.5 to 0.0). The original run's decoding produced a constant answer on one item (Q12). Under the training-format readout the model predicts the majority answer for most respondents there and its mean probabilities follow the true shares, so the collapse came from the decoding, not the adapter.

TABLE IV. PEW FOLD 0: 46,209 HELD-OUT ANSWERS FROM 6,000 RESPONDENTS. NLL: NEGATIVE LOG-LIKELIHOOD; ECE: EXPECTED CALIBRATION ERROR.

| Method | Accuracy (%) [95% CI] | NLL | ECE |
|---|---|---|---|
| Majority class | 30.4 [29.9, 30.8] | 1.59 | 0.001 |
| Logistic regression | 38.5 [38.0, 39.0] | 1.47 | 0.004 |
| Gradient boosting | 40.0 [39.6, 40.5] | 1.44 | 0.005 |
| Fine-tuned Qwen2.5-7B | 38.0 [37.5, 38.4] | 1.48 | 0.009 |
| Zero-shot Qwen2.5-7B (raw) | 18.0 [17.6, 18.4] | 3.07 | 0.452 |
| Zero-shot Qwen2.5-7B (chat) | 16.1 [15.7, 16.4] | 4.26 | 0.519 |

On the 200-respondent subset the fine-tuned model is 20.2 points above the zero-shot model (paired 95% CI 16.9 to 23.5) and 11.4 points above the prior-corrected one. Its group differences correlate r = 0.58 with the truth on the subset (GBM 0.64) and 0.82 on the full fold (GBM 0.91). The slope is near one on the subset (1.07) and above one on the full fold (1.87, GBM 1.85). The two use different cell thresholds (15 and 100 respondents), so they are not directly comparable. Hard predictions inflate between-group differences for all three methods (2.0 times for the fine-tuned model, 2.0 for GBM, 2.6 for logistic regression). That is consistent with argmax predictions inflating group differences in general, though we did not test soft predictions.

The fine-tuned model's subgroup gaps (Table V) differ from zero on five of eight axes after Holm correction: religion 10.0 points, region 8.8, age 2.7, income 1.7 and urban/rural 1.3. They differ from GBM's by at most 1.9 points, nominally only for region (+1.4, CI 0.1 to 3.5, uncorrected). On probabilities, its predicted subgroup distributions are close to the truth. The mean Jensen-Shannon divergence is 0.009 across religion cells and 0.006 across region cells, against 0.004 and 0.002 for logistic regression, 0.029 and 0.020 for the item marginal, and 0.271 and 0.293 for the zero-shot model. So the fine-tuned model is two to three times further from the truth than the classifiers on religion and region, and more than ten times closer than the zero-shot model. Tuning the classifiers on a validation split of the training respondents changes little (logistic 38.5%, GBM 40.2%), so the gap to GBM is not due to under-tuned classifiers. The fine-tuned configuration itself was not tuned.

TABLE V. SUBGROUP ACCURACY GAP (BEST MINUS WORST CATEGORY, PERCENTAGE POINTS) ON PEW FOLD 0, WITH THE PERMUTATION P-VALUE FOR THE FINE-TUNED MODEL AND ITS DIFFERENCE FROM GBM'S GAP.

| Axis | Majority | GBM | Fine-tuned gap [95% CI] | p | Fine-tuned minus GBM [95% CI] |
|---|---|---|---|---|---|
| Religion | 5.3 | 9.1 | 10.0 [7.3, 13.5] | <0.001 | +1.0 [-2.1, +4.0] |
| Region | 13.4 | 7.4 | 8.8 [7.1, 11.3] | <0.001 | +1.4 [+0.1, +3.5] |
| Education | 6.6 | 1.2 | 2.0 [0.9, 3.6] | 0.058 | +0.8 [-0.8, +2.2] |
| Age | 5.6 | 3.8 | 2.7 [1.7, 4.2] | 0.005 | -1.2 [-2.4, +0.5] |
| Caste | 2.8 | 1.5 | 3.4 [1.5, 6.3] | 0.071 | +1.9 [-1.5, +4.0] |
| Urban/rural | 2.9 | 2.0 | 1.3 [0.3, 2.2] | 0.012 | -0.7 [-1.5, +0.1] |
| Income | 2.6 | 1.5 | 1.7 [0.6, 2.7] | <0.001 | +0.2 [-0.6, +1.1] |
| Sex | 2.1 | 0.7 | 0.4 [0.0, 1.3] | 0.37 | -0.3 [-1.0, +0.5] |

## C. RQ3: No Detectable Gain on Unseen Items

We applied the adapter to eight Pew items it never saw. On the trained items the MLX re-implementation scores 32.9% against 18.8% for its base (a gain of 14.1 points), and the original model scores 40.5%. The port therefore reproduces much of the effect but is weaker; row agreement is 56% and directional validity is similar (0.71 against 0.63 for the original). On the unseen items (Fig. 3) the adapter scores 28.9% and its base 28.5%, a difference of +0.4 points (-5.1 to +5.9), against +14.1 in domain. Over respondents the in-domain gain exceeds the unseen gain by 13.7 points (95% CI 7.0 to 20.4), but that interval treats the eight items on each side as fixed. Across items the difference is 14.0 points with a standard error of 10.2 (permutation p = 0.18), and the unseen gain runs from -21 to +36 points per item. We therefore do not claim the in-domain gain is reliably larger. The safe reading is that no gain on unseen items is detectable. Two weaker signals remain: calibration (NLL 2.39 to 1.69, still above the uniform value of 1.65) and the direction of group differences (r = 0.24, 95% CI 0.03 to 0.35, against -0.02 for the base). Classifiers fitted with labels on these items reach 37.6% (majority), 44.7% and 47.6%, so here the adapted LLM is no substitute for data on the new question. The test rests on the approximate port, 120 respondents, eight items and one narrow training set, so it does not show that broader fine-tuning would also fail to transfer.

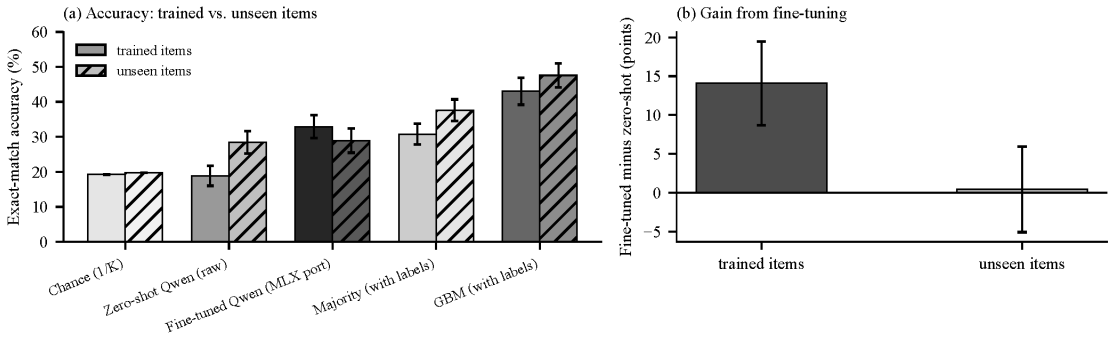



Fig. 3. (a) Accuracy on the eight trained and eight unseen items (120 respondents; 95% respondent-bootstrap CIs). (b) Gain of the fine-tuned model over its zero-shot base, with paired 95% CIs. Classifiers marked "with labels" are not available for a genuinely new question.

## D. RQ4: Scale and Data

**Larger zero-shot model.** gpt-oss-120b scores 27.6% (24.9 to 30.3) on 100 respondents (774 rows). That is 2.3 points below the majority class (30.0%, not significant), against 18.7% for the 7B model on the same rows, and 11.8 points below the fine-tuned 7B model (7.1 to 16.5). It is a different, larger model and we did not vary scale within one family, so we cannot credit the difference from the 7B model to scale. It reaches the majority class and goes no further.

**Labelled data.** Classical models need a fair amount of labelled data before they beat the majority class. Averaged over five random subsets, logistic regression matches the majority class up to N = 100 respondents (30.5% against 29.1%), beats it by 3.9 points at N = 400 (33.7%) and by 6.4 at N = 1,600 (36.4%). GBM clears the majority class in accuracy only at about 1,600 respondents (34.6%), and its NLL is still worse than the majority class's there (1.69 against 1.59). On the same 200 test respondents, the fine-tuned 7B model, trained on 23,999 respondents, scores 37.7%, slightly above logistic regression trained on 1,600 (36.4%); the gap is within the sampling error of 200 respondents. Whether LLM adaptation is more sample-efficient than classical models with little data is the open question most likely to favour LLMs. We could not test it reliably with 7B models on our hardware, and a small-model (0.5B) attempt on up to 100 respondents collapsed to a near-constant answer, so we draw no conclusion from it.

## E. WVS-7 India Replication

Across the 144 screened items the majority class reaches 45.5% and the cell mode 45.4%, against 46.9% for logistic regression and 50.1% for GBM, so demographics add only 1.4 and 4.6 points. On the 15 evaluated items, Gemini 3.1

Flash Lite (30 respondents) scores 25.7% and gpt-oss-120b (40) scores 23.0%. Both are within sampling error of the majority class and 9.5 and 11.3 points below GBM. Zero-shot Qwen2.5-7B (120 respondents, 1,716 rows) scores 22.1%, 2.7 points below the majority class (95% CI -5.6 to +0.1) and 15.3 below GBM. Its group differences show no directional validity ($r = -0.03$ over 345 cells; GBM 0.64), the profile changes accuracy by -1.3 points (-4.2 to +1.5), and on seven of the 15 items (nine at a 90% threshold) it gives the same answer to nearly every respondent. The 15 items come from an earlier run and have their own majority class of 21 to 25%, well below the 144-item figure, so each LLM is compared with the majority class on the same rows. Subgroup gaps cannot be told apart from random group assignment in any of 12 API-model tests (11 of 12 with $p > 0.05$, none after Holm correction) or in the 120-respondent run (smallest $p = 0.082$). Detection thresholds range from 4.5 to 26.7 points, above most gaps seen on Pew (1 to 2.5 points).

## VII. Discussion

**What the profile does.** A zero-shot model does not ignore the profile. It varies with it far more than real answers do, in directions that match the data only weakly, and the net effect on exact-match accuracy is slightly negative. That fits small, noisy persona effects [5] and the over-determination reported in [3]. We did not vary prompts, so we cannot say whether richer or better-worded demographics would help.

**What fine-tuning does.** Fine-tuning on respondent-level data gets item marginals and group differences close to a classifier's, but not all the way. Accuracy ends 2.1 points below GBM and 0.5 below logistic regression, directional validity on the full fold is 0.82 against 0.91, and the two models agree on 72% of predictions. The absence of a detectable gain on unseen items fits an item-specific mapping from profile to answer. Given the port, the number of items and the narrow training set, it is not evidence that transfer is impossible.

**Where LLM respondents could add value.** On this evidence, not as substitutes for labelled data on the same question, since matched classifiers are cheaper and more accurate. Nor, at 7B with narrow fine-tuning, as predictors for new questions. The open settings are the ones our design cannot test: richer text-conditioned personas, much broader fine-tuning across many items, and aggregate distribution estimates from probabilities rather than argmax.

**Cost and benefit.** Section V shows what each answer costs: about 0.2 s of T4 time per row for the 7B model, hours of GPU time to fine-tune, and seconds for a classifier. Where matched classifiers are equally accurate on the trained questions and the adapter shows no detectable gain on new ones, the cost favours the classifier. The case for LLM respondents would have to rest on settings classifiers cannot serve, which the held-out-item test probed only weakly.

**Evaluation practice.** Accuracy comparisons need a majority-class baseline (45.5% over the 144 WVS-India items). Inflation and concentration of hard predictions should be read against baselines that show them too. Subgroup-gap claims need a permutation null and detection thresholds, which at tens of respondents exceed any plausible effect.

## VIII. Limitations

Everything comes from one fold, one seed and one epoch. Scoring used fp16 on a T4 while training used bf16, and the original run's greedy decoding of the same adapter scores lower (36.1%), so results depend on the readout. We use the one that matches training and the zero-shot conditions. Fine-tuning used eight religion- and fertility-related items and a full-sequence loss, so transfer to broader item sets is untested and the unseen-item null may reflect narrow training. The transfer and probability results use an MLX re-implementation of the adapter that is about 7 to 8 points weaker than the original on the same rows. The transfer comparison is therefore conservative for the original model, and it rests on eight items and 120 respondents with large item-level uncertainty. The local zero-shot controls use random subsets of 100 to 200 respondents, though the full-fold zero-shot result agrees. We tested one 7B model locally and two free-tier API models and did not test frontier models on these surveys (Chen et al. [3] did, on US and cross-national data, and saw the same failures). We used two surveys with eight and 15 items and no language axis. Pew survey weights are not applied, so statements describe the sample, not the population. Accuracy is exact match on ordinal items with argmax predictions, directional-validity cells share respondents across axes, and the variance share is an in-sample increase in R-squared.

## IX. Conclusions and Future Work

A prompted 7B model varies strongly with a demographic profile but uses it weakly and slightly harmfully. Fine-tuning on respondent-level data recovers most, though not all, of the accuracy and direction gap to classifiers on the trained questions, and shows no detectable accuracy gain on eight new ones. In our settings matched classifiers remain the benchmark LLM respondents have to beat, and they are far cheaper to run.

Future work has four parts. First, fine-tune on many items and test transfer on dozens of unseen items with an item-level bootstrap, using the original adapter instead of a port. Second, repeat the study across folds and seeds. Third, test richer text-conditioned personas and aggregate distribution estimates from calibrated probabilities. Fourth, measure the cost side more carefully, including smaller and distilled models and energy use. Code, prompts and analysis scripts are at github.com/DevilHunter695/silicon-sampling-capstone. WVS-7 must be obtained from the WVS Association and the Pew data from Pew Research Center.

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